

CryptoCopilot — Demo Day

Speaker Script

Friendly, simple English (B1–B2) · about 10 minutes · one section per slide.

Read it out loud a few times — the bold words are the ones to stress.

Total speaking time ≈ 10 minutes · 12 slides

How to use this script

- Each section matches one slide. The grey line shows what is on the screen.
- **Bold** words are the key points — say them slowly and with energy.
- Replace **[your name]** on slide 1. You don't need to memorise word-for-word — keep the order and the bold points, and speak naturally.
- The small time tag (e.g. ~55s) is a guide, not a rule. If you run long, the data and results slides are the easiest to shorten.

Slide 1 · Title — CryptoCopilot (~20s)

On screen: Title slide: the project name and the disclaimer.

Hi everyone, I'm [your name], and this is **CryptoCopilot**. It's a smart assistant for people who are **new to the crypto world**.

In the next ten minutes, I'll show you what it does, how I built it, and what it can — and can't — do. One quick note first: this is **not financial advice**, and it never touches real money. So, let's start with a simple question: who really controls your money?

Slide 2 · The hook — money, control, and a different idea (~55s)

On screen: Two cards: traditional money vs crypto.

Think about normal money, like euros or dollars. A **central bank** controls it. Your money sits in a bank, and the bank makes the rules. You even pay **fees** just to hold and move your own money. Someone else is always in charge.

Crypto is a different idea. It is decentralized. That means no single boss and no central bank. You hold your own money directly. That gives real freedom — it's open and global.

But it also has downsides. It's complex, the prices jump up and down a lot, and for a beginner it's honestly **confusing**. So that's the problem I wanted to solve. A newcomer looks at the crypto market and just sees noise. **CryptoCopilot turns that scattered data and news into clear information and simple signals** about each coin.

Slide 3 · Architecture — one system, two languages (~55s)

On screen: The architecture diagram: five containers, one database.

Let's look under the hood. CryptoCopilot is built from **five small parts**, called containers, around **one shared database**.

At the front is the website you see, built with **React**. It talks to the **backend** — the brain — written in **Java with Spring Boot**. The backend does the technical analysis, the AI chat, the Analyst, and the trading. On the left, a **Python** service does the machine learning, and a second Python service lets me run those jobs on demand.

Here's the key idea, and the part I'm most proud of. The two languages, Python and Java, **never call each other directly**. They only share one database. **Python writes; Java reads**. It's simple, clean, and very reliable.

Slide 4 · The data — 10 coins, 5 free sources (~50s)

On screen: Coin list + two cards of data sources.

Good signals need good data. CryptoCopilot follows the **ten biggest coins** — Bitcoin, Ethereum, Solana, and so on.

The data comes from **five free, public sources**. Binance gives the prices. CoinGecko gives market and community data. Five news sites give headlines. And two more give **on-chain** activity — what's really happening on the blockchain.

In total, that's about **two hundred and thirty thousand rows** of data in one database. And it's built to be safe: if any source goes down, the system just **skips it and keeps going**. It never crashes.

Slide 5 · Machine learning — an honest edge (~55s)

On screen: The ML pipeline strip + a signal card.

Now the machine learning. This is the heart of the project.

I start with raw price candles. From them I build **forty-six features** — useful numbers that describe the market. Then I train a model called **XGBoost**. It learns from about two years of history to predict one thing: will the price go **up, down, or stay flat** over the next twenty-four hours?

Two things I really care about. First, the probabilities are **calibrated** — when it says seventy percent, it really means about seventy percent. Second, a tool called **SHAP** explains the top reasons behind every prediction.

And I'm honest about the score. Its accuracy, called **AUC, is zero point five-eight**. That's a **small but real edge** above pure luck. No small model can beat the market, and I don't pretend it does.

Slide 6 • Technical analysis — signals you can audit (~45s)

On screen: The Signals page.

The model is just one opinion. So I added a second, very different one: classic **technical analysis**.

Using a Java library called **ta4j**, the backend reads the same price charts and applies famous indicators — **Ichimoku, RSI, MACD**, and more. From clear rules, it gives a simple verdict: **bullish, neutral, or bearish**.

And the best part — it shows you **exactly which rules fired**. Nothing is hidden. You can check every step yourself.

Slide 7 • The Researcher — AI that cites its sources (~55s)

On screen: The Analyst page with cited headlines.

Next, the part that makes it feel smart: the **Researcher**. It's a chat you can ask questions.

Normally, AI chatbots **make things up**. I didn't want that. So I use a method called **RAG**. In simple words: before the AI answers, it **searches my real data** — news, on-chain data, and a knowledge base. Then it answers using **only those sources**, and it shows the **citations**.

If the answer isn't in the sources, it just says so. And it **never gives trading advice**. The nice thing: by default it runs **for free, on my own computer**. But with one click, I can switch to a stronger OpenAI model.

Slide 8 • The Analyst — four views, one opinion (~45s)

On screen: The Analyst page, full view.

So now I have **four different views** on each coin: the machine learning signal, the technical analysis, the fundamentals, and the news.

The **Analyst** brings them together into **one clear opinion** — a direction, how strong it is, and how much the four views agree.

It's fully rule-based, so it's predictable and you can **audit** it. And a safety check makes sure the summary only uses **real numbers** — it can't invent things.

Slide 9 • Paper trading — practice with no real money (~45s)

On screen: Paper Trades + Performance pages.

What if you want to **act** on a signal? You can — safely. CryptoCopilot has **paper trading**. That means real trading mechanics, but with **fake money**. Nothing is ever at risk.

You place an order, and the system fills it realistically, with a small fee, just like a real exchange. Then it **tracks your account** and shows your performance — your equity curve and your risk numbers.

And again, I keep it honest. The simple test strategy **doesn't make money**. The point was to build the engine **correctly**, not to get rich.

Slide 10 · The app — clean and simple (~40s)

On screen: Markets + ML Pipeline screenshots.

All of this lives in a clean **web app**, built with React and TypeScript, so it's fast and reliable.

There are **seven pages** — markets, signals, the Analyst, the chat, your trades, and performance. There's even a page where I can run the **whole pipeline** — get the data, train the model, make predictions — right from the browser, with one click.

And every single page reminds you: this is **decision-support, not financial advice**.

Slide 11 · Honest results (~45s)

On screen: Three result cards: ML, RAG, engineering.

Let me be straight about the results.

The machine learning has a **small, real edge** — an AUC of zero point five-eight. The AI search finds the right source **ninety percent** of the time, and always with citations. And the whole thing — five services, three languages — is **tested and automated**.

My goal was never to beat the market. My goal was to build a **real, production-grade system, end to end**, and to report the numbers **honestly**. The rules I never broke: **paper money only**, and always **decision-support, not advice**.

Slide 12 · Thank you (~15s)

On screen: Closing slide.

And that's **CryptoCopilot** — four views on every coin, joined into **one clear, explainable opinion**. Thank you so much for listening. I'd love to take any questions.